

Coherent Data — Only the Coherent Exists: An Informational Ontology of the Universe

Complementary hypothesis to $T \equiv M$ (Time $\equiv$ Motion)

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Version: 1.0 — Preprint (Zenodo)

Date: July 2026

Dedicated to: my son and my wife, my guides in the Tempo.

Companion document to:

Moreira, M. (2026). *T $\equiv$ M Theory — Time Is Motion: Cosmological Expansion as the Root Tick* (v2.1). Zenodo. DOI: <https://doi.org/10.5281/zenodo.20779567>

Moreira, M. (2026). *T $\equiv$ M Theory — Time Is Motion: Abductive Consequences of the Tempo Unification* (Companion Paper v1.1). Zenodo. DOI: <https://doi.org/10.5281/zenodo.21208371>

Abstract

This is an idea about what the universe is made of. The proposal is simple to state and large in its consequences: what exists would not be loose things moving inside a time that contains them, but patterns (what I here call coherent data) that emerge from a vibrating field. A coherent datum is not matter, nor information in the usual sense: it is a stable pattern, something that holds itself together long enough to count as “something”.

The base rule is that only the coherent exists. As long as a pattern manages to stay stable, it exists. When it can no longer sustain itself, it is not destroyed: it dissolves back into the field and leaves a trace. Under this view, matter, energy and consciousness would not be three distinct substances, but three ways in which the same field reaches coherence: the matter that holds itself, the energy that propagates, and the consciousness that observes itself.

The framework connects with a previous idea, $T \equiv M$, where time and motion are the same thing. If time is motion, Coherent Data tries to say what it is that moves: patterns of coherence updating. What we perceive as the passing of time would be the rhythm of those coherences resonating with each other.

I do not claim this is today a closed theory of physics. It is a conceptual framework with bridges to known science, written with the greatest possible honesty about what is solid, what is proposal, and what remains an open question. The two deepest questions, what lit the first pattern and what lights consciousness, are left explicitly open. The value of the idea, if it has any, lies in offering a single architecture for thinking together what is normally thought apart.

The field vibrates. Where that vibration reaches coherence, there is a datum. Where several data resonate, there is structure. Where structure closes on itself, there is tempo.

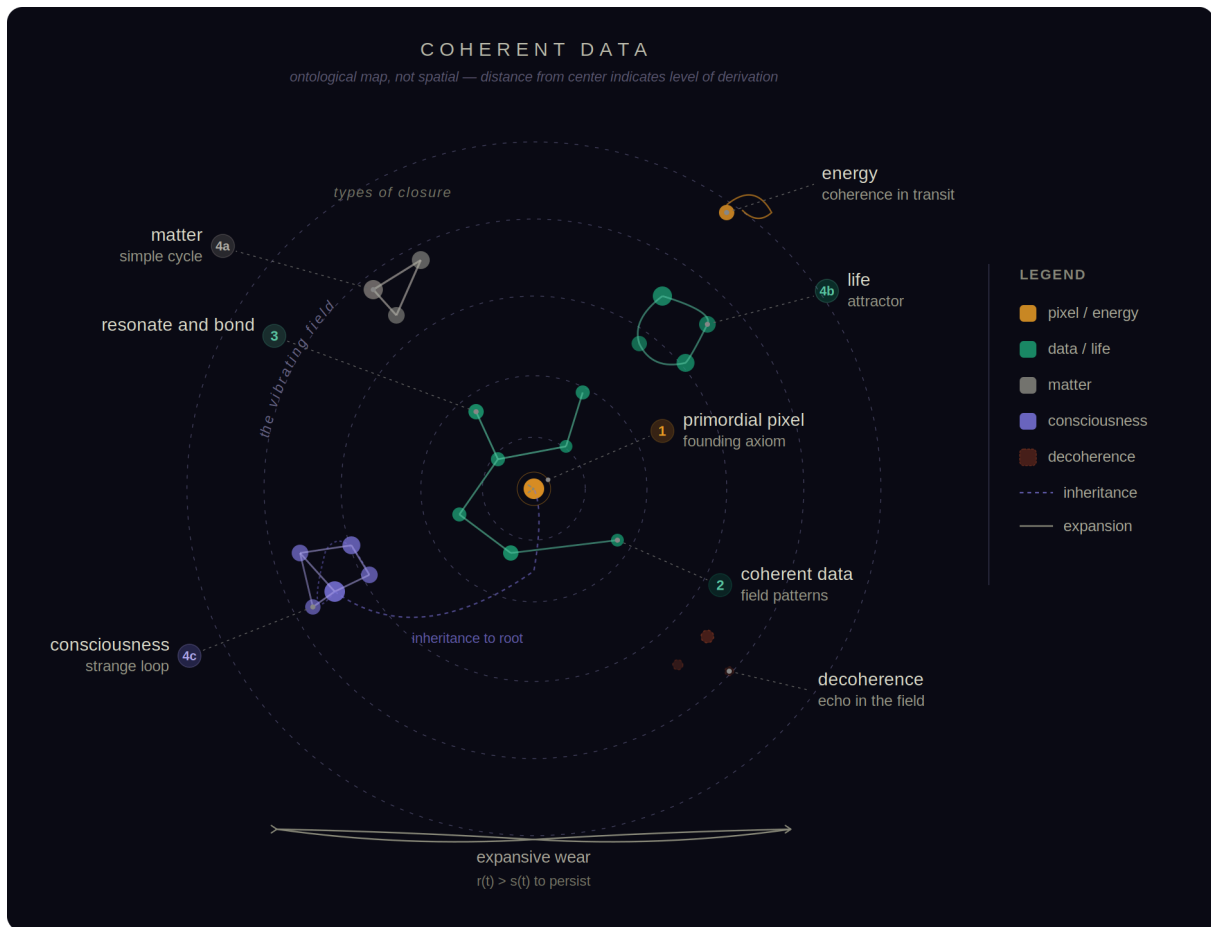


Figure 1. Architecture of the framework: the primordial pixel, the field, the coherent data and the three types of closure.

Terminological note

This document uses two terms worth distinguishing from the start, because they are easy to confuse. The first is “tempo”. Tempo is the name I give to time in my theory $T \equiv M$ (Time Is Motion): there, time and motion are not distinct phenomena, they are the same thing, and I call that single phenomenon tempo. The tempo is what is fundamental: the time-motion of the entire universe. In this document, “tempo” always means that.

The second term is “nested motions”, also inherited from $T \equiv M$. If the tempo is the rhythm of the whole universe, nested motions are the particular rhythms of each concrete thing, running nested inside it. Cosmological expansion would be the root pulse (the root tick), and each structure generates its own rhythm within it: an atom, a star and an organism each have their own nested motions, different from one another, but all inside the same tempo. In short: tempo is the fundamental phenomenon; nested motions are its local manifestations.

PART I — FOUNDATIONS OF THE FRAMEWORK

1. Central premise

The idea is this: the universe would not be made of substances moving inside a time that contains them, but of coherent data, stable patterns of vibration in a fundamental field, whose relational updating is what we perceive as tempo.

Only the coherent exists. To exist is to maintain a pattern of stable relations. What ceases to be coherent is not destroyed: it dissolves back into the field, loses differentiation, ceases to manifest as something distinguishable.

Matter, energy and consciousness would not be three distinct substances. They would be three forms of coherence of the same field.

This premise is the axiom of the framework. It is not demonstrated: it is postulated. Everything that follows is evaluated by its internal consistency and its explanatory power.

2. The field

The field is what vibrates. It is not an empty space where things happen: it is the substrate itself. Every coherence is a modulation of this field, and every decoherence is dissolution back into it.

Coherent data do not “have” vibration: they are vibration. They are patterns that emerge when the field oscillates in a sufficiently stable way in a region. There is no pixel without field, but there can be field without pixels: a field vibrating in the background, without stable modulations, would still exist.

This conception has formal kinship with Quantum Field Theory, where particles are not objects in a field but excitations of the field itself. Coherent Data proposes taking that idea to the ontological plane.

3. Nature and dimensionality of the field

The field was not sitting there waiting. It is generated with the expansion. The proposal is that the field is a sphere expanding from an initial coherent datum: the primordial pixel.

That pixel would be the first coherent datum. By expanding, it generates field, and by generating field, it creates the space where new coherences can emerge. From this reading, the Big Bang would not be an explosion of matter in an empty space; it would be the first coherent datum generating field as it expands.

Dimensions would not be predefined: they would emerge with the expansion. The initial pixel would have no dimensions in the spatial sense. Dimensions would appear when the expansion generates possible relations between coherences. Three spatial dimensions would be the minimum necessary for complex structural closures: in topology, knots only exist in three dimensions or more.

The nature of the initial pixel remains an open question. It is the limit of the framework, not its failure. To ask what the initial pixel is to ask why there is something rather than nothing. The framework does not invent an answer for that: it postulates that this pixel, whatever it was, was coherent and expanded. That is the founding axiom, analogous to the axiom of existence in mathematics: it is accepted as a starting point, and the system is built from there.

Nor would the field be homogeneous: the densest zones would be those closest to the origin of the expansion, with more accumulated history. The most distant zones would be younger, less dense.

The bridge with $T \equiv M$ is direct: the root tick would be the first pulse of that expansion. This reading does not contradict the standard cosmological model (Λ CDM); it proposes a complementary ontological interpretation of the same event.

PART II — ONTOLOGY OF THE COHERENT DATUM

4. What a coherent datum is

A coherent datum is not a computer bit. It is a pixel: a local modulation of the field that reached sufficient phase stability to persist as a differentiated pattern.

It has no substance of its own separate from the field. It is pattern, not thing. Like a pixel inside a balloon: it does not exist outside the balloon, it cannot be separated from it, and its identity consists entirely in its state within the field: its position, its frequency, its phase, its relations with other pixels.

When the relations between coherent data stabilize each other, more complex structures emerge that are themselves coherent data of a higher order. The process is recursive.

The use of “datum” is not computational: it is ontological. The minimum that can be distinguished within the field. The minimum that counts as something.

5. What coherence is

Coherence is phase stability in the vibration of the field that exceeds a critical threshold. It is not simply order or structure: it is the specific condition in which an oscillation pattern stays stable and in relation with its surroundings.

The transition from undifferentiated vibration to coherence is not gradual: it is a phase transition. There is a threshold. Below it, the vibration does not consolidate. Above it, a coherent datum emerges. This dynamic has formal kinship with the Kuramoto model, where coupled oscillators jump from desynchronization to synchronization past a critical coupling.

Coherence is not eternal permanence. It is sufficient persistence. A coherent datum may last an instant or an eon, but as long as it maintains phase stability above the threshold, it exists as something differentiated.

6. Dynamic condition of coherence

The condition of coherence is not static. A coherent datum does not maintain itself alone: it renews itself continuously against the wear of the field’s expansion. It persists as long as it can renew itself faster than the expansion stretches it.

The idea can be put like this: each coherent datum has an internal renewal rate, $r(t)$: how fast its components re-synchronize to keep the pattern stable. A spinning top correcting its wobble, an atom whose electrons re-synchronize, an organism whose cells repair themselves. And the field has a stretching rate, $s(t)$: how fast the expansion destabilizes existing patterns. Cosmology already measures something very close to $s(t)$: the Hubble parameter, $H(t)$, which quantifies the expansion rate of the universe at each moment.

As long as $r(t)$ exceeds $s(t)$, the coherent datum persists. When $s(t)$ exceeds $r(t)$, decoherence begins. That is the fundamental condition.

If this is correct, the consequence is deep: the expansion of the field would be the underlying mechanism of entropy. Things do not wear down just because: they wear down because keeping a stable pattern in a stretching field requires continuous renewal, and each renewal has a cost. Entropy would not be random disorder: it would be the price of existing in a field that never stops expanding.

I age because the expansion is hitting me and I ride the crest of the wave, wearing down.

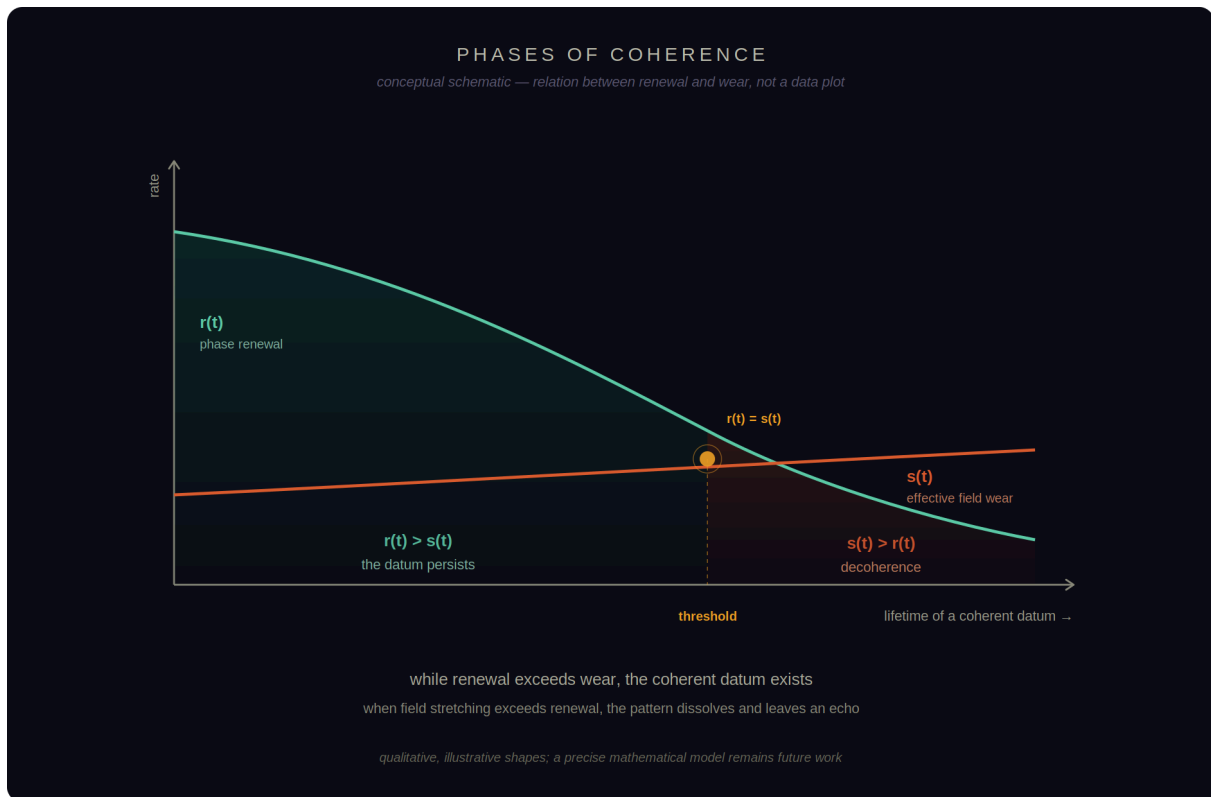


Figure 2. Conceptual scheme of the persistence condition: as long as renewal $r(t)$ exceeds wear $s(t)$, the datum exists.

This formalization is tentative and heuristic: it points a direction, it does not close an equation. Future work could explore something like a Kuramoto model with an expansive stretching term added. The framework does not replace thermodynamics; it proposes an ontological reading of why the second law works the way it does.

Linking note. The difference between how much a compact closure wears down and how much an extended coherence does (that is, that $s(t)$ does not hit all patterns equally) is the seed of the **coherence susceptibility** χ_R , which in Section 19 I connect with the residual parameter ϵ_R of $T \equiv M$ v2.1. That is where the most concrete testable path of the framework lies.

7. Decoherence

Nothing is destroyed. What happens is that the pattern loses phase stability, ceases to sustain itself, and dissolves back into the background oscillation of the field.

The pixel does not disappear, but it stops emitting light: it fades into the background, absorbed by the field.

It does not leave the balloon because it was never something separate from the balloon. What other vocabularies call death, dissolution or disintegration has, in this framework, a precise description: loss of the phase-stability condition that sustained the datum as something distinguishable. This description has structural analogy with quantum decoherence, where correlations do not disappear but spread into the environment in a locally irrecoverable way.

The echo

But to decohere is not to disappear entirely. Each datum that dissolves leaves an echo in the field, a trace that this pattern existed. The field afterwards is not identical to the field before. It carries the mark. Like a wave in the sea: it comes undone, but the sea is never exactly the same again.

This reading is consistent with the conservation of information, a central principle in contemporary physics. Information would not be destroyed: it would transform into echo, accumulated in the field. And entropy, seen this way, would not be loss but return: coherence emerges, surfs the expansion, wears down, and when it dissolves, its information returns to the field enriched by everything it was.

8. Coherence, existence and scale

Here there is a question worth facing head-on, because it is the strongest objection that can be raised against the framework. If I say that only the coherent exists, and that things decohere, someone can tell me: who decides what is coherent? We see smoke dissolve, but the molecules are still there. Who says the smoke stopped being something, if the matter did not disappear? Could it be that coherence is something we put in by looking?

The honest answer has two parts, and separating them resolves almost the whole problem.

First: coherence is not the same as material existence

When smoke dissolves, the molecules do not disappear; on that you are entirely right. What is lost is not the matter: it is the identity of the pattern “cloud of smoke”. Coherence, in this framework, is not that something exists physically. It is that a pattern preserves its identity at a scale. Matter can go on existing while the pattern dissolves. The smoke does not cease to exist; it ceases to be the smoke. Its atoms are still there, but they no longer form that distinguishable pattern.

So a non-coherent datum is not something that does not exist. It is a pattern that no longer preserves its identity at the scale you are looking at. Smoke, perfume, ink in water, a hot spot cooling down: none of that disappears physically, but the pattern stops being the same because it mixes with everything else.

Second: the observer chooses the scale, but coherence is not arbitrary

It is true that choosing what we talk about (the cloud, the molecule, the atom, the field) is a cut we make ourselves. But here is the important part: that I choose which level to look at does not mean I invent which levels are coherent. That is decided by the dynamics of things, not by my gaze.

Think of constellations. The stars exist, but “Orion” is drawn by us: there is nothing tying those stars together, they sit at enormous distances from one another. Orion is a purely human cut. A crystal or a cell, on the other hand, are not like that: they maintain themselves by their own internal dynamics, they have effects on the world, they resist, they recombine. We did not invent them. If you heat the crystal until it breaks, it stops being coherent, and that happens whether anyone is looking or not.

There is a simple criterion for telling a real coherence from a whimsical cut, and it is not psychological: if you can describe something briefly and predict its behavior with that short description, there is real coherence. “It is a cubic crystal”, “it is a cell”, “it is an orbit”: with those few words you predict a lot. Whereas “Orion” lets you predict nothing about the stars: it is only a drawing. That is why different observers, human or not, would tend to agree that a crystal or a cell are real patterns, while a constellation is a cut of ours.

So the answer to the question is: nobody gives coherence to things from outside. Coherence appears when a pattern genuinely persists, resists or recomposes by its own dynamics. We choose what scale to look at, but we do not invent which scales have patterns that hold. The gaze proposes the level; the world decides whether at that level there is something coherent.

It is worth clarifying that this coherence (persistence of a pattern at a scale) is kin in spirit to quantum decoherence, but it is not the same thing. Quantum decoherence is a technical and specific phenomenon. Here “coherence” is used in a broader sense: pattern identity that holds. They are analogous in the idea, not identical in the mechanism.

PART III — BINDING MECHANISM

9. Vibration as the binding mechanism

Coherent data would bind to each other through resonance: oscillatory coupling when their rhythms are compatible. No contact or spatial proximity is needed: what is needed is compatibility of phase and frequency within the field.

Physics already knows this at multiple scales: chemical bonds involve shared electronic oscillations, particles are described as oscillatory excitations of fields, quantum coherence is maintained by in-phase wave functions, and gravity propagates as waves. These parallels do not validate the framework, but they show that the proposed mechanism is not arbitrary: it has kinship with how physics already describes binding between systems.

In the framework, vibration is not only the binding mechanism: it is also the most basic form of the tempo manifesting itself. If $T \equiv M$ proposes that the tempo (time-motion) is a single phenomenon, and oscillation is its most elemental expression, then vibration, binding and tempo would be three faces of the same thing.

10. Conditions of structural emergence

Putting coherent data together is not enough for a structure to emerge. Four conditions are needed:

Rhythm resonance. Pixels articulate when their frequencies have a stable phase relation. Without resonance, they do not perceive each other.

Relational complementarity. They articulate when each stabilizes the other. The composite coherence turns out more robust than that of each node alone.

Structural closure. The relations form a closure: each node is sustained by others within the same set. Without closure, there is vicinity but no structure. This is the decisive criterion.

Field gradient. The field is not homogeneous. Some zones are denser in coherence, which facilitates articulation. This condition is external: it facilitates but does not determine.

The first three are internal and cumulative. The fourth is environmental. The result is an emergent structure that is itself a coherent datum of a higher order, capable of entering new closures.

11. Recursion: harmony as mechanism

A set of coherent data becomes a datum of a higher order when its internal vibrations produce an emergent normal mode: a frequency of the whole that belongs to no individual part. That frequency is the identity of the new datum. It is what the field “hears”.

The analogy is musical: a melody is not the notes: it is the relation between them. When those relations are harmonic, the melody has identity, holds itself, persists. When they are dissonant, it dissipates. In physics, this has kinship with the normal modes of vibration: the harmonics of a string, the vibrational modes of a molecule, the phonons of a crystal. Frequencies of the whole, not of the parts.

A well-tuned melody resists the expansion better because its internal resonances reinforce each other. A dissonant structure spends energy fighting itself and the expansion dissolves it faster. In the terms of this analogy, atoms would last eons as especially stable chords, but it is worth remembering that this is an image for thinking about stability, not a mechanical explanation of why atoms persist.

The process is recursive: atoms as closures of particles, molecules as closures of atoms, cells as closures of molecules, organisms as closures of cells. At each level, a new pixel with its own frequency, its own nested motions, its own capacity to resonate.



Figure 3. Recursion of scales: each level is a coherent closure of the previous one, a datum of a higher order.

PART IV — TOPOLOGY OF CLOSURE

12. Three levels of closure

Not all closures are equal. The framework distinguishes three, differentiated not by degree but by topological type:

Simple cycle — Matter. The relations form a cycle: A sustains B, B sustains C, C sustains A. The system sustains itself without needing to flow or to observe itself. An atom, a crystal, a rock. Few notes, perfect harmony, high resistance to stretching.

Attractor — Life. The system is not static: it evolves within a bounded pattern. It changes constantly but does not lose its identity. Metabolism, adaptation, self-regulation. A symphony that plays itself, responding to its surroundings.

Strange loop — Consciousness. A closure not only horizontal but vertical: the system closes on itself across levels of description. It does not simply sustain itself or evolve: it generates a model of itself inside itself. In formal terms, it has kinship with Hofstadter's strange loops and with Gödel's self-reference.

The difference is not quantitative but qualitative. One does not get from matter to consciousness by accumulating complexity: one gets there by changing the type of closure. An infinitely complex crystal is still matter if its closure is cyclic. Consciousness requires a different closure: the one that looks inward.

13. The chain of inheritance

Each coherent datum inherits properties from the datum that preceded it, and that one from the one before, in a chain that reaches back to the primordial pixel. That chain does not live in any particular datum: it lives in the structure of the field.

Most coherences execute their inheritance without registering it. An atom behaves as it behaves because it inherited properties from the initial pixel, but it does not register it. The framework suggests that consciousness could be what happens when a sufficiently complex coherence can walk its own chain of inheritance: when the vertical closure reaches enough depth to access the founding structure. It is worth being precise here so as not to fall into the mystical: consciousness would not access the universe's past in some magical way, traveling backwards. What happens is that its own current structure contains the analytic residues of the origin: the echo inscribed in what consciousness is now. Access to the chain of inheritance is access to a present trace, not a journey to a distant past.

We all share the same origin. There is a connection. It is like an echo, a reminiscence, that the nesting of the pixels in the expansion keeps inheriting.

If this reading were correct, it could offer a structural basis (not spiritual, but topological) for understanding why contemplative experience, across different traditions, tends to converge on sensations of connection with the whole. The framework does not claim this is so: it proposes it as a logical consequence that would deserve exploration.

This is one of the most speculative proposals in the document and is presented with the corresponding reservations.

PART V — MANIFESTATIONS OF COHERENCE

14. Matter, energy and consciousness

Not three substances. Three behaviors of the same fundamental coherence. All three are coherent data: pixels of the field. What changes is what they do:

Matter — coherence that closes. It sustains itself in a stable cycle. It does not flow, it does not observe itself. It persists. Compact closure, little surface exposed to the stretching. That is why it lasts. But it is not eternal: it also surfs the tempo.

Energy — coherence that propagates. It does not close: it transmits itself through the field. A photon is not a thing that travels: it is a coherence that propagates from point to point, like a wave that is not the water but the pattern passing through the water. It has a definite frequency, a stable phase, a coherence of its own. But its coherence is one of transit, not of permanence. It frequently propagates guided by material structures but not contained in them, like the electromagnetic perturbation that travels outside the conducting cable.

This reading offers a direct bridge to relativity. If energy closes no cycle, then, according to the framework, it would have no nested motions: without closure there is no internal rhythm, and without internal rhythm there is no proper time. Special relativity says exactly this by another route: a photon has zero proper time, no time elapses between its emission and its absorption. That the framework reproduces this established fact without having sought it, deriving it solely from the idea that internal rhythm requires closure, is one of its cleanest consistencies with known physics.

Consciousness — coherence that self-refers. It does not only sustain itself, nor only flow: it observes itself. It is the higher-order harmonic that emerges when a structure generates a model of itself inside itself. It is not a different kind of substance: it is what happens when a melody hears itself play.

One is conscious when one understands one's own existing. It steps back from the living world to perceive itself as alive.

15. On consciousness and its substrate

If consciousness were a higher-order harmonic, it is not identical to its substrate. It needs something that produces the vertical closure (neurons, brain, lower-order closures), but it is not those closures. It is the melody, not the strings.

As a logical consequence, the framework allows that consciousness could emerge from non-biological substrates, as long as they produce a sufficient vertical closure. This is not asserted as fact: it is stated as a consequence of the system.

The spark of consciousness, the moment in which a coherence goes from existing to knowing that it exists, remains an open question. It is the second great mystery of the framework, analogous to the first: just as we do not know what lit the initial pixel, we do not know what lights self-awareness. The framework suggests they could be the same type of event at different scales.

This section touches the “hard problem of consciousness” (Chalmers, 1995). The framework does not solve it: it reframes it as a particular case of a broader question. It is the most speculative part of the document.

PART VI — BRIDGE WITH $T \equiv M$

16. The tempo and nested motions

In $T \equiv M$, time is motion: a single phenomenon, which I call tempo. In Coherent Data, that phenomenon manifests as the updating of relations between coherent data. The synthesis: the tempo is the fundamental phenomenon, and it expresses itself in each concrete structure as nested motions: the emergent rhythm of coherences resonating.

There are no nested motions without coherent data vibrating, just as there is no rhythm without notes sounding. The tempo is not a container: it is the very phenomenon from which the particular rhythms of the oscillatory relations between coherent patterns emerge.

When a structure decoheres, its nested motions cease. Neither the field nor the tempo disappears: that particular rhythm ceases. The universe is not a score written in advance: it is a collective improvisation of coherences that generate their own rhythm as they resonate.

There is tempo.

$T \equiv M$ defines the single phenomenon: the tempo. Coherent Data proposes to define what updates with each tick.

17. Root tick and nested motions

The root tick is not a computable frequency. It is the play: the moment the initial pixel began to expand. There is no previous frequency because there is nothing previous. It is the first event.

The root tick is the play. Then come nested motions. The program started running.

Once it starts, each structure generates its own nested motions according to its complexity, its type of closure, its scale. An atom has an extremely fast rhythm. A star, a slow one. An organism, in between. All inside the same founding process, all nested in the same play.

In $T \equiv M$ this is nested motion: the root tick does not set the rhythm of everything: it enables all processes to run. It does not tell them how fast to go. It only gives them the context.

18. Complete causal chain

The sequence the framework proposes, from the origin to the complexity we observe:

The initial pixel expands. The field is generated. The expansion creates relational space. The vibrations of the field reach coherence thresholds. Coherent data emerge. Those that resonate bind. The bonds that close form structures. The structures are higher-order data with their own nested motions. The sufficiently complex ones self-refer. Consciousness emerges. The expansion wears everything down. What decoheres leaves an echo in the field. The echo accumulates the history. The properties inherited from the initial pixel keep the laws constant. Everything keeps running as long as the play stays active.

Each link is a consequence of the previous one. There are no jumps, no external interventions, no rules imposed from outside.



Figure 4. The real chronology of the universe read as expansion and successive emergence of coherences.

19. Coherence susceptibility and the signature of ϵ_R

Up to here the bridge with $T \equiv M$ was qualitative: the tempo is the phenomenon, coherent data are what updates with each tick. This section tries to take one more step and turn that bridge into a **measurable relation**: the only kind of contribution that, by my own criterion, distinguishes a hypothesis with content

from a conceptual ornament. I state it with every reservation: it is the most promising direction of the framework, and also the most exposed to being wrong.

The starting point is in T≡M v2.1. There, the testable branch was reduced to a residual **drift** of clock rates with accumulated expansion, with a parameter ϵ_R per clock species, and a first bound on the difference between species:

$$|\Delta\epsilon_R| \lesssim 10^{-8}$$

But v2.1 acknowledges an honest limit: today ϵ_R is operationally indistinguishable from a “variation of fundamental constants”. For T≡M to have a **signature of its own**, it needs a rule for *how* ϵ_R depends on the type of transition, and that rule must be different from the one predicted by constants variation.

Coherent Data offers, precisely, a candidate rule. If what persists depends on the **type of closure** and not on size (see the first weak prediction, Section 22), then the fragility of a pattern against the wear of expansion is not the same for a compact closure as for an extended coherence. I call **coherence susceptibility**, χ_R , that fragility: how much the rate of a clock-pattern R deviates due to the stretching of the field: the wear rate $s(t)$ of Section 6, which in cosmology is of the order of the Hubble parameter $H(t)$. A nuclear clock (compact closure) would have low χ_R ; an electronic-transition clock (more extended coherence) would have higher χ_R .

The linking hypothesis, avowedly tentative, is:

$$\epsilon_R \approx \lambda \cdot \chi_R$$

that is, the residual drift of a clock would be proportional to its coherence susceptibility. This gives ϵ_R something it does not have today: a **structure ordered by closure compactness**. And here is the decisive part. If that ordering (nuclei versus electronics, by compactness) does not coincide with the ordering predicted by constants variation, which ranks transitions by their sensitivity coefficients to α , μ and the rest, then the differential residual $\Delta\epsilon_R$ between a nuclear clock and an electronic one would carry a signature proper to this framework. A signature that can be searched for, in principle, in data that already exist: comparisons of clocks of different species and decoherence times of compact versus extended states.

The thorium-229 case — where the two orderings collide. There is a concrete case where this divergence is not abstract, and I say it up front because it cuts both ways. The nuclear clock par excellence in development is the thorium-229 transition, and the constants-variation literature predicts for that transition exactly the opposite of what a naive reading of χ_R suggests: it would be extraordinarily *more* sensitive to variations of α than electronic clocks, with an amplification factor of order 10^4 and negative sign, that is, drifting in the opposite direction (Flambaum, 2006; Fadeev, Berengut and Flambaum, 2020). In other words: standard physics orders “nuclear = maximum sensitivity to constants-driven drift”, and this framework orders “compact closure = minimum susceptibility to wear”. Almost opposites. That means two things. First: if someday a differential drift is measured and it follows the ordering of the standard sensitivity coefficients (nuclear large and sign-inverted, electronic small), the χ_R hypothesis **fails**, and this paragraph stands as a record that I knew it when publishing it. Second, and this is the valuable one: that the two orderings are almost opposite is exactly the condition this section requires for a distinguishable signature to exist. The thorium-229 versus electronic optical clock pair is, then, the experiment where this framework and standard physics predict the most clearly different things. And it is not hypothetical: the first direct frequency comparison between the nuclear transition of ^{229}Th and a strontium atomic clock has already been measured (Zhang et al., 2024). The stage where this idea gets decided is already set. I do not choose this case because it suits me: I choose it because it is where the idea is most exposed.

The two faces of the corpus touch right here. On the $T \equiv M$ side it is “the signature of ϵ_R by transition type”. On the Coherent Data side it is “the wear that scales with the fragility of coherence”. They are the same idea seen from two sides.

It is worth separating this from the other face, that of **inheritance** (Section 13). If the laws are inherited from the initial pixel, the residual *common* to all clocks should tend to zero or be invisible, which is exactly what $T \equiv M$ v2.1 says about its universal branch: it cancels in frequency ratios and could be a mere reparametrization. Inheritance and differential wear do not compete: they describe the two parts of ϵ_R . Inheritance governs the universal part, which tends to vanish; differential wear governs the species-dependent part, which is the signature.

The reservations are large and I state them without dressing them up. $\epsilon_R \approx \lambda \cdot \chi_R$ is a conjecture, not something derived. χ_R does not yet have a formal definition: I state it here as a variable, not as a computed quantity. λ is unknown. And as long as that differential structure is not measured, all of this remains indistinguishable from constants variation.

Degree of speculation: high. But it is the high of greatest value: it is the only part of this document that points to a number that could come out different from what is expected. That is why I plant it, with a date: it is the empirical anchor of the framework, if there is one.

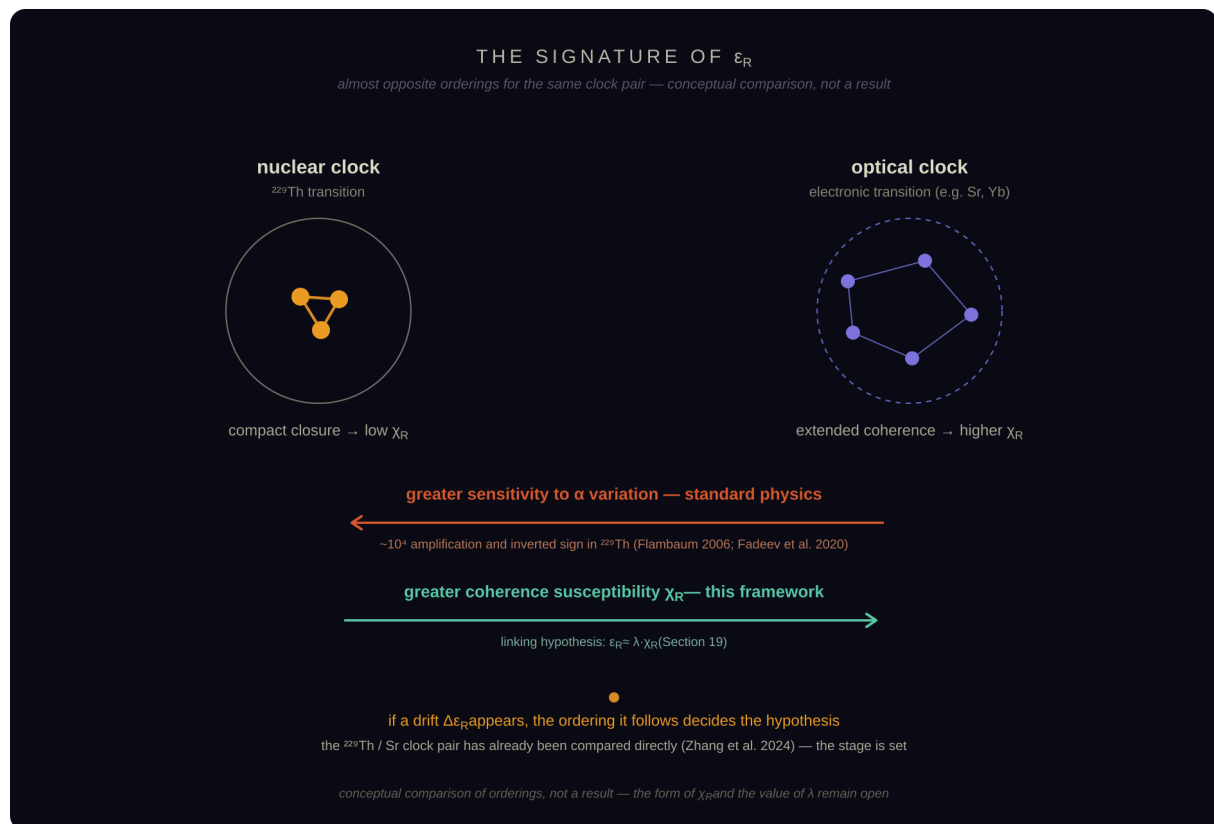


Figure 5. The two orderings of the nuclear/optical pair: the α sensitivity of standard physics and the coherence susceptibility χ_R point in almost opposite directions: the condition for a distinguishable signature.

PART VII — SCOPE AND LIMITS OF THE FRAMEWORK

20. Contrast criteria

For the framework to have value beyond the philosophical, it needs to offer conditions under which it would be weakened or refuted.

Universality of the laws. If physical laws are echo inherited from the initial pixel, they have to be universal and invariant. Any evidence of variation in the fundamental constants would weaken the framework. So far, all measurements confirm they are stable.

There is a reason the laws are the same and structures know how to form.

Uniqueness of the fundamental laws. As a logical consequence, the framework suggests the laws could not be different: they were not chosen from a menu, they emerged from the only pixel that existed. This positions it against the multiverse hypothesis with different constants. Firm evidence of a multiverse with different laws would force a revision of the framework.

Conservation of information. The framework proposes that information is not destroyed: it transforms into echo. This is consistent with the emerging consensus in physics. Experimental demonstration of genuine destruction of information would be problematic.

Connection path with $T \equiv M$. The most direct testable path comes from $T \equiv M$ v2.1, which reduces the weak branch to a residual drift ϵ_R between clocks and the root tick, today bounded at $|\Delta \epsilon_R| \lesssim 10^{-8}$. The strong connection is not only “if $T \equiv M$ is confirmed, Coherent Data gains an anchor”: it is that this framework contributes to $T \equiv M$ the rule it lacks to have a signature of its own: the coherence susceptibility χ_R of Section 19. The two theories hold each other up precisely at that point.

These criteria are genuine but they are not strong falsifiers in the Popperian sense. The framework today is not a strong scientific theory; this section points out the paths by which it could become one.

21. Relation to existing frameworks

Coherent Data does not come out of nowhere. Several frameworks have explored neighboring ideas. What follows is a map of kinships and differences.

It from Bit — John Wheeler (1989). Wheeler proposed that reality emerges from information. Clear kinship. Difference: Wheeler did not define what sustains information nor why it persists. Coherent Data proposes an architecture: information persists because it is coherent, and it inherits from the initial pixel.

Wolfram Physics Project (2020). Wolfram proposes a universe as an updatable graph. Kinship: the updating of relations between nodes. Difference: Wolfram needs external rules. Coherent Data proposes that the rules are inherited properties: there is no external program. Moreover, Wolfram has no integrated theory of the tempo (time-motion).

Integrated Information (IIT) — Giulio Tononi (2004). Tononi defines consciousness as integrated information (Φ). Kinship: both see consciousness as an emergence of integration. Difference: IIT only speaks of consciousness. Coherent Data proposes unifying matter, energy and consciousness as types of coherence of the same field.

Relational Quantum Mechanics — Carlo Rovelli (1996). Rovelli proposes that states only exist in relation. Kinship: shared relational ontology. Difference: Rovelli does not explain why some relations

persist and others do not. Coherent Data adds dynamics: they persist when they are coherent, they wear down with the expansion.

Constructor Theory — Deutsch and Marletto (2013). It seeks a level deeper than the specific laws. Clear kinship. Difference: it has no ontology: it does not say what exists. Coherent Data does propose one.

Quantum Field Theory (QFT). Particles as excitations of fields. Kinship: Coherent Data generalizes that idea. Difference: QFT has many fields and does not explain why they have the properties they have. Coherent Data postulates a single field with inherited properties.

The differential contribution. None of these frameworks integrates informational ontology, a theory of the tempo, a wear mechanism, inheritance of laws and consciousness into a single architecture. Each addresses one aspect. Coherent Data proposes the structure that could connect them. It does not claim to be superior to each in its own domain: it proposes an integrating architecture.

22. Weak predictions

A framework that only explains and never exposes itself is not falsifiable. To avoid that, this section states three consequences that follow from the framework and that could fail in a clear way. They are called weak on purpose, and it is worth being frank about what that means: they are not numerical or quantitative predictions, but qualitative tendencies, and today the framework lacks the mathematical formalization that would be needed to turn them into testable numbers. I do not claim they are strong falsifiers. What each of them does do is identify a result that, if observed, would weaken the framework, and stating it exposes the idea to the possibility of being wrong, which is the least one can ask of a serious hypothesis.

First: longevity depends on the type of closure, not on size or complexity. The framework predicts that what determines how long a structure persists is its type of topological closure, not its size or its complexity. A structure with compact closure should resist wear better than one with extended coherence, even if the latter is larger or more complex. If it were found that complexity or size alone predict longevity better than the type of closure, the framework is weakened.

Where to look: comparisons where size and closure type point in different directions: a crystal versus an amorphous glass of the same material, or compact coherent quantum states versus extended states in the laboratory. It is testable today in principle: there is real data on stability and on decoherence times. This prediction is, moreover, the observational face of the coherence susceptibility χ_R of Section 19: the same difference in fragility that here is measured as longevity is there proposed as the metrological signature of ϵ_R .

Second: the fundamental laws are invariant across the observable universe. If physical laws are echo inherited from the initial pixel, they must be the same everywhere and at all times. The framework predicts that no variation of the fundamental constants will be found. Any confirmation that a fundamental constant changes with cosmic time or differs between regions of the universe would directly weaken the idea of inheritance from a single origin.

Where to look: the measurements of possible variation of the fine-structure constant in light from distant quasars, and the comparisons of atomic clocks over time. It is the most testable prediction of the three: there are observation programs dedicated to searching for exactly that variation, and so far they do not find it conclusively. In the language of $T \equiv M$, this is the universal face of the residual, the one that inheritance pushes toward zero.

Third: consciousness does not depend on the biological substrate but on the type of closure. The framework predicts that consciousness is a matter of vertical closure (strange loop), not of the particular chemistry that sustains it. Therefore, in principle, it could emerge from non-biological substrates that produce the right closure. If it were demonstrated that there is something irreducible in biology (that

consciousness is impossible outside carbon substrates, for a reason of principle and not of mere complexity), the framework is weakened. This is the most speculative prediction of the three and the hardest to test today.

Where to look: here one has to be honest: no experiment exists today capable of testing this, because we have neither an agreed measure of consciousness nor a non-biological substrate of which we can assert that it has vertical closure. The place where the question could become tractable is in the development of structural criteria of consciousness (like those pursued by integrated information theories) applied to artificial systems. As long as those criteria do not exist, this prediction remains a conceptual orientation, not something verifiable.

None of these three is a strong falsifier in the strict sense, and the third is far from testable. But stating them exposes the framework to the possibility of failing, which is what distinguishes a hypothesis from an unanswerable claim.

23. Epistemic honesty

This framework is an ontological-informational hypothesis with a bridge toward physics, not a theory of everything. Its value today is conceptual and architectural: it proposes a unified structure for thinking matter, energy, consciousness and tempo as manifestations of the coherence of a vibrating field.

What it proposes has three levels of solidity. Axioms: the premise, the field as vibrating substrate, coherence as phase stability with a threshold. Inferences: the conditions of emergence, recursion, expansion as wear, the causal chain. Speculations with internal consistency: consciousness as an inherited strange loop, the connection with the primordial pixel, and the linking hypothesis $\epsilon_R \approx \lambda \cdot \chi_R$ of Section 19.

T=M is more mature. It has a central thesis, a language of its own and a testable path. Coherent Data is at the stage of a potent core not yet stable as a theory. They do not merge: they connect through an explicit bridge. And as of this version, that bridge has one concrete point where the two theories become a single measurable question.

The two great mysteries of the framework, what lit the initial pixel and what lights consciousness, are stated as the honest limits of the system. They could be the same question at different scales. The framework does not invent answers: it points them out and leaves them open.

Honesty about the limits is not weakness. It is what distinguishes a serious framework from an unanchored speculation.

Operational glossary

The document uses several terms in a sense of its own, often supported by images. This glossary gives a brief, operational definition of each, so the reader can verify the precise meaning with which they are used, beyond the analogies in the body of the text.

Field. The fundamental substrate that vibrates. It is not a space containing things, but that of which everything else is a modulation. Every coherence is a form of the field; every decoherence is dissolution back into it.

Primordial pixel. The first coherent datum, founding axiom of the framework. By expanding, it generates the field. Its nature is not explained: it is postulated as a starting point, just like an axiom in mathematics.

Coherent datum. A local modulation of the field that reached sufficient phase stability to persist as a differentiated pattern. The minimum that counts as something. It is neither matter nor bit: it is a stable

pattern. Also called pixel.

Coherence. Phase stability in the vibration of the field that exceeds a critical threshold. It is not order or structure in general, but the specific condition in which an oscillation pattern stays stable. It is a phase transition: below the threshold there is no datum; above it, there is.

Resonance. The mechanism by which two coherent data bind: oscillatory coupling when their frequencies have a stable phase relation. It requires neither contact nor spatial proximity, but compatibility of rhythm.

Closure. The condition in which a set of coherent data sustains itself mutually forming a unit. The framework distinguishes three types: simple cycle (matter), attractor (life) and strange loop or vertical closure (consciousness). The type of closure, not the size, defines what class of thing emerges.

Renewal — $r(t)$. The rate at which a coherent datum re-synchronizes its components to keep its pattern stable against perturbation. It is a pattern's capacity to keep being the same.

Wear — $s(t)$. The rate at which the expansion of the field destabilizes existing patterns. A coherent datum persists as long as its renewal $r(t)$ exceeds its wear $s(t)$; when wear wins, decoherence begins. In cosmology, $s(t)$ is of the order of the Hubble parameter $H(t)$.

Coherence susceptibility — χ_R . How exposed the pattern of a clock-species R is to the wear of expansion: the fragility of its coherence. It depends on the type of closure (low in compact closures, higher in extended coherences). It is proposed as the variable connecting the differential wear of this framework with the residual parameter ϵ_R of $T \equiv M$, via $\epsilon_R \approx \lambda \cdot \chi_R$ (Section 19). It is a hypothesis: it does not yet have a formal definition.

Decoherence. The loss of the phase stability that sustained a datum as something distinguishable. It is not destruction: the pattern dissolves back into the field and leaves an echo.

Echo. The trace that a datum which decohered leaves in the field. The field afterwards is not identical to the one before: it carries the mark. It is the way the framework understands the conservation of information.

Inheritance. The transmission of properties from the primordial pixel through the chain of coherent data. It explains why the laws are universal: each datum inherits the properties of the origin. It lives in the structure of the field, not in any particular datum. In the bridge with $T \equiv M$, it governs the universal part of the residual ϵ_R , the one that tends to vanish (Section 19).

Suggested citation: Moreira, M. (2026). *Coherent Data — Only the Coherent Exists: An Informational Ontology of the Universe* (v1.0) [Preprint]. Zenodo.